LLM with Construction Engineering Prior Knowledge via a Knowledge Graph

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***Abstract*—On large construction projects, engineering instructions and method statements must travel through a long chain of members—from the client to the site engineer, the foreman, and finally the workers on site. In practice this chain is lossy: information is fragmented across chat applications and paper records, critical numeric tolerances in repair procedures are misread, and Thai–English language barriers with a multinational workforce introduce further ambiguity, all of which translates into rework and non-conformance reports. We present SmartCivil, a question-answering framework that couples a large language model (LLM) with a knowledge graph to give site members reliable, source-grounded access to engineering knowledge. Scanned bilingual method statements are ingested by a vision-language OCR model; the extracted text, headings, tables, and figures populate a knowledge graph whose entities and relations make the engineering knowledge explicit and queryable. For each user question the system performs hybrid retrieval—keyword, full-text, and dense semantic search—whose rankings are merged by reciprocal rank fusion, and an LLM synthesises an answer that is constrained to the retrieved context and annotated with page-level citations. We evaluate the framework on a tunnel-segment repair corpus using both single-category and multi-category questions, scoring engineering-answer correctness, document grounding, bilingual fidelity, and response latency. The results characterise where retrieval-augmented generation succeeds and where document grounding remains the limiting factor for safety-critical construction question answering.**

***Index Terms*—large language models, retrieval-augmented generation, knowledge graph, optical character recognition, hybrid retrieval, reciprocal rank fusion, bilingual question answering, construction informatics**

I. Introduction

Large language models (LLMs) have become strong general-purpose reasoners over natural language, yet when deployed as domain question-answering systems they exhibit a well-known failure mode: they hallucinate, producing fluent answers that are not supported by any authoritative source. In safety-critical engineering domains, where a single misquoted tolerance can invalidate a repair procedure, ungrounded generation is unacceptable. Retrieval-augmented generation (RAG) mitigates this by conditioning the model on passages retrieved from a trusted corpus, so that generation is anchored to verifiable evidence rather than parametric memory.

The effectiveness of RAG is bounded by the quality of its retrieval stage. Sparse lexical retrieval such as BM25 rewards exact term overlap and is robust for rare technical identifiers (for example, repair-method codes), whereas dense retrieval over learned embeddings captures semantic similarity and paraphrase. Because the two signals are complementary, we adopt a hybrid retriever and combine its component rankings with reciprocal rank fusion (RRF), a rank-based aggregation that requires no score calibration across retrievers. Beyond flat passage retrieval, a knowledge graph imposes explicit structure on the corpus—entities such as defects, materials, and procedures, together with the relations among them—providing a grounding substrate that supports multi-hop and constraint-bearing queries that plain chunk retrieval handles poorly.

Two further technical challenges arise from the source material. First, the engineering knowledge exists only as scanned, image-based method statements, so the corpus must be reconstructed with optical character recognition (OCR) using a vision-language model that can recover text, headings, and tabular structure. Second, the documents and the user population are bilingual (Thai and English) and frequently code-switched, which requires retrieval and generation to preserve technical terminology across languages. We further employ query decomposition for compound questions and constrain the generator to emit structured, citation-bearing answers so that every claim is traceable to a source page.

This paper makes the following technical contributions:

- a bilingual document-ingestion pipeline that converts scanned method statements into a structured, retrievable knowledge base via vision-language OCR;
- a hybrid keyword, full-text, and dense retriever whose rankings are fused by RRF and consumed by an LLM under a grounding constraint with page-level citation; and
- a multi-dimensional evaluation protocol—covering engineering-answer correctness, document grounding, bilingual fidelity, and latency—applied to a tunnel-segment repair corpus.

II. Methodology

Building on the RAG-LLM design of POLRAG [1], SmartCivil is organised as a six-phase pipeline that transforms scanned, image-based method statements into grounded, citation-bearing answers. The phases are document ingestion, knowledge-graph construction, query understanding, retrieval, answer generation, and structured formatting.

A. Document Ingestion

The source corpus consists of scanned bilingual (Thai–English) method statements for tunnel-segment repair. Each page is processed by a vision-language OCR model that recovers running text, headings, tables, and figures while preserving the page index. The output is serialised into a manifest together with extracted image and table assets, so that every text segment retains a back-pointer to its originating page; these page references are later surfaced as citations and are the basis of the document-grounding evaluation.

B. Knowledge-Graph Construction

The ingested content is intended to populate a knowledge graph whose nodes are domain entities (defects, materials, procedures, and quantitative conditions) and whose edges encode the relations among them, providing explicit structure for multi-hop and constraint-bearing queries. *The detailed construction of this graph is deferred to future work; the experiments reported here operate over the segmented document store produced in the ingestion phase.*

C. Query Understanding

An incoming question is first analysed for scope. Compound questions that span multiple categories are decomposed into atomic sub-questions, each of which is routed through retrieval independently before the partial results are recombined. This decomposition is what allows the system to answer multi-category questions that pair, for example, a repair-method query with an environmental-measure query.

D. Retrieval

For each (sub-)question the system runs three retrievers in parallel. A keyword retriever rewards exact matches on rare technical identifiers; a full-text retriever operates over both Thai and English surface forms; and a dense retriever ranks segments by the cosine similarity between the question embedding $\mathbf{q}$ and each segment embedding $\mathbf{d}$,

$$\mathrm{sim}(\mathbf{q},\mathbf{d}) = \frac{\mathbf{q}\cdot\mathbf{d}}{\|\mathbf{q}\|\,\|\mathbf{d}\|}. \tag{1}$$

The three rankings are merged by reciprocal rank fusion (RRF), which aggregates on rank position and therefore needs no score normalisation across retrievers,

$$\mathrm{RRF}(d) = \sum_{r\in R} \frac{1}{k+\mathrm{rank}_r(d)}, \tag{2}$$

where R is the set of retrievers, $\mathrm{rank}_r(d)$ is the rank of segment d under retriever r, and k is a smoothing constant. The top-ranked fused segments form the context passed to the generator.

E. Answer Generation and Structured Formatting

A system prompt constrains the LLM to answer strictly from the retrieved context, to decline when the context is insufficient, and to respond in the language of the question. The raw answer is then passed through a formatting step that filters OCR noise and emits a structured response: the textual answer, any supporting image or table evidence, and the page-level document citation.

III. Experiment Design

We evaluate two complementary aspects of the system: the quality of the retrieval stage in isolation, and the end-to-end quality of the generated answers.

A. Semantic Similarity Retrieval

Retrieval is assessed with context precision and context recall over the retrieved segment set. With true positives (TP) being retrieved segments that contain correct and relevant information with respect to the expected answer, false positives (FP) being retrieved segments that contribute no relevant information, and false negatives (FN) being relevant segments that were not retrieved, the metrics are

$$\text{Context Precision} = \frac{TP}{TP+FP}, \qquad \text{Context Recall} = \frac{TP}{TP+FN}. \tag{3}$$

Precision measures how much of the retrieved content is actually relevant, while recall measures how completely the retriever covers the evidence needed to answer the query.

B. LLM Performance

End-to-end answer quality is scored on a 100-point rubric spanning five dimensions, summarised in Table I. Engineering-answer correctness and document-grounding accuracy carry the largest weights because they capture, respectively, whether the prescribed repair method matches the method statement and whether the answer is faithful to the source documents without hallucination.

TABLE I
LLM Performance Evaluation Rubric

Evaluation Dimension	**Max**	**Metric / Method**
Engineering Answer Correctness	40	Expert validation, rule matching
Document Grounding Accuracy	30	Citation correctness, hallucination rate
Bilingual Capability (TH–EN)	10	Translation accuracy, semantic consistency
Response Latency	10	Average response time (s)
System Prompt Compliance	10	Prompt adherence score
Total	**100**	

Each scored dimension is graded into qualitative bands. For engineering-answer correctness, an answer is *Excellent* (36–40) when it identifies the correct repair method in line with the method statement (e.g. RP-TN-01, RP-TN-02, or Reject) together with the complete numeric conditions such as crack width, depth, and void size; *Good* (26–35) when the method is correct but some numeric detail is missing; *Fair* (16–25) when the intent is partially correct but the chosen method departs from the guideline; and *Poor* (0–15) when the method is wrong or contradicts the documents. For document-grounding accuracy, an answer is *Excellent* (26–30) when it is entirely supported by the corpus with no hallucination, *Good* (18–25) when citations are largely correct with minor over-summarisation, and *Poor* (0–17) when it contains clear hallucination or unsupported claims. Response latency is mapped from the mean per-question time: ≤ 5 s scores 10, 6–10 s scores 8, 11–20 s scores 5, and > 20 s scores 0.

IV. Results and Discussion

We evaluate the fused-retrieval configuration on two question sets: a single-category set (Level 1, $n = 20$) and a multi-category set that pairs two or three topics per question (Level 2, $n = 20$). Each answer was scored against an expert ground truth on the rubric of Table I; latency was derived from the measured per-question time, while bilingual capability and system-prompt compliance were assessed at the system level. The aggregate scores are reported in Table II.

TABLE II
Evaluation Results (Fused Retrieval)

Dimension	**Max**	**L1**	**L2**
Engineering Answer Correctness	40	19.6	22.4
Document Grounding Accuracy	30	19.4	20.6
Response Latency	10	0.0	0.0
Bilingual Capability (TH–EN)	10	9.0	9.0
System Prompt Compliance	10	9.0	9.0
Total	**100**	**57.0**	**61.0**

A. Latency

Every question in both sets exceeded the 20 s threshold (Level 1 mean 35.7 s, Level 2 mean 43.7 s, worst cases above 80 s), so response latency scores zero throughout. This is the single largest and most actionable deficit, driven by the sequential per-mode generation and the second answer-reformatting pass.

B. Retrieval Recall vs. Hallucination

Recall, rather than hallucination, is the dominant failure mode. In 9 of 20 Level 1 questions and 7 of 20 Level 2 questions the system returned “information not found” even though the answer was present in the corpus—a retrieval miss rather than a generation error. When relevant context *was* retrieved, answers were typically accurate and faithful, yielding eight Excellent (Level 1) and seven Excellent (Level 2) correctness grades. Only one genuine hallucination was observed, where a project-name query returned a long list of unrelated projects.

C. Citation Quality

Page-level grounding is capped by noisy citations: correct answers frequently returned five candidate pages where only one was relevant, and several otherwise-correct answers cited the wrong page entirely. Multi-category (Level 2) questions scored slightly higher overall because their compound phrasing supplied richer retrieval cues, but they commonly answered the repair-method sub-question correctly while missing a second numeric sub-part such as a mixing ratio or pot life.

D. Bilingual Fidelity and Compliance

Bilingual capability and prompt compliance are strong: answers consistently matched the language of the question and preserved English material and technical terms, and the frequent honest abstentions indicate good adherence to the instruction to answer only from retrieved context. Overall, the framework is reliable when retrieval succeeds; improving retrieval recall and citation precision, and reducing latency, are the priorities for future work.

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References

[1] H. Lin, P. Deng, Q. Zhong, and X. Zhu, “POLRAG: A RAG-LLM framework for policy question answering,” in *Proc. 2025 IEEE Int. Conf. High Performance Computing and Communications (HPCC)*, 2025, pp. 1259–1264, doi: 10.1109/HPCC67675.2025.00179.